

UNIVERSITY NAME

*Rigid Body Mechanics — Final Exam**Year: 2014***Exercise 1:**

Let $R_0(O, \bar{x}_0, \bar{y}_0, \bar{z}_0)$ be a Galilean orthonormal direct frame relative to which we propose to study the motion of a system (Σ) consisting of:

- A straight rod (T) along the direction class $(O, \bar{x})$, with negligible cross-section and mass, which can rotate freely around its fixed endpoint O. - A solid homogeneous sphere (S) of mass M , radius a , and center G.

The sphere (S) is pierced along its axis $(G, \bar{x})$ by the rod (T). It can slide along the rod and rotate around it.

All friction is neglected (i.e., the different contacts are considered frictionless) and we assume the connection at O is perfect. The motion parameters of (Σ) in the frame R_0 are the three usual Euler angles (ψ, θ, ϕ) and $\overline{OG} = \rho(t)$.

All results will be expressed in the basis $(i, \bar{w}, \bar{z})$.

1. Determine the following instantaneous rotation vectors:

$$\hat{\Omega}(T/R_0), \quad \hat{\Omega}(S/T), \quad \hat{\Omega}(S/R_0)$$

2. Calculate the velocity and acceleration vectors of point G relative to R_0 .
3. Calculate the inertia matrix at point O of the system Σ :

$$\Pi_0(\Sigma)$$

4. Calculate the angular momentum about O of the system Σ relative to R_0 :

$$\tilde{L}_0(\Sigma/R_0)$$

5. Show that:

$$\tilde{L}_0(\Sigma/R_0) \cdot \tilde{z}_0 = \text{const} = k_1 \quad \text{and} \quad \tilde{L}_0(\Sigma/R_0) \cdot \tilde{z} = \text{const} = k_2.$$

Deduce two equations of motion in the form of first integrals.

6. Determine the kinetic energy of the system Σ relative to the frame R_0 .
7. By applying the kinetic energy theorem to Σ , give a third equation of motion in the form of a first integral. (Assume the rod-sphere connection is perfect.)

8. In the special case where the axis $(O, \bar{z})$ remains coincident with the axis $(O, \bar{z}_0)$, determine $\rho(t)$. Given:

$$\rho(0) = 0 \quad \text{and} \quad \dot{\rho}(0) = \rho_0$$

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$$\tilde{L}_0(\Sigma/R_0) \cdot \tilde{z}_0 = \text{const} = k_1 \quad \text{and} \quad \tilde{L}_0(\Sigma/R_0) \cdot \tilde{z} = \text{const} = k_2.$$

Deduce two equations of motion in the form of first integrals.

9. Determine the kinetic energy of the system Σ relative to the frame R_0 .
10. By applying the kinetic energy theorem to Σ , give a third equation of motion in the form of a first integral. (Assume the rod-sphere connection is perfect.)
11. In the special case where the axis $(O, \bar{z})$ remains coincident with the axis $(O, \bar{z}_0)$, determine $\rho(t)$. Given:

$$\rho(0) = 0 \quad \text{and} \quad \dot{\rho}(0) = \rho_0$$

System Diagram

Answer Area

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.
- 11.